



2025 Data Mining

Final Project: Knowledge Tracing Prediction



Task Introduction

Goal : For each student's interaction with a question, predict whether they will answer correctly.

- You are free to use any model or approach for this task.
- Requirement
 - Upload your submission to Kaggle
 - Submit a report and your source code to E3



Dataset

- train.csv
 - Student interactions and responses for training
- test.csv
 - Student interactions for prediction
- questions.json
 - Question metadata and content information
- concept.json
 - Knowledge concept mapping



Training Data

train.csv - each row contains student interaction data with the following columns:

- uid - Integer. Unique student identifier.
- question_id - Integer. Identifier for the question being answered.
- concept_id - Integer. Knowledge concept identifier referencing concept.json.
- response - Binary. Student's response correctness (1 for correct, 0 for incorrect).
- timestamp - Integer. Unix timestamp indicating when the response was recorded.



Testing Data

test.csv - each row contains student interaction data with the following columns:

- uid - Integer. Unique student identifier.
- question_id - Integer. Identifier for the question being answered.
- concept_id - Integer. Knowledge concept identifier referencing concept.json.
- timestamp - Integer. Unix timestamp indicating when the response was recorded.

Note: test.csv does not include the 'response' column as this is what needs to be predicted.



Kaggle

- [Competition Link](#)
- Create a team with your **group ID** , we use this information for grading. Use [this link](#) to find your group ID, your team name should be `group_<groupID>` .
- If you failed to do so under any circumstances, there will be **a penalty of 5 points** to your score, so be sure to use the correct team name.



Kaggle (cont.)

- Public leaderboard is calculated with 50% of the test set, private leaderboard is calculated with the other 50%, the final standings may be different.
- DO NOT **overtune** your model to fit the public leaderboard, or you will suffer from overfitting
- The scoring metric is **AUROC** .



Kaggle Submission

- Report user with their index (column “uid” in test.csv) and the corresponding prediction to each answer.
- There should be **3614 x 2** entries in your csv file, with columns “uid” and “response” .
- Please ensure your model outputs continuous probability values between 0 and 1 (**not binary**). This approach better reflects the model's confidence in each prediction.
- The order of the ids does not matter. Refer to sample_submission.csv for the correct format.



Report Submission

Answer the 3 questions. **Please answer the questions in detail to receive full points for each question.**

1. (10%) What preprocessing and feature engineering methods did you apply to the knowledge tracing data? Additionally, please discuss how different feature representations (e.g., question embeddings, concept embeddings, temporal features) contributed to model performance based on your experimental results.



Report Submission (cont.)

2. (10%) Describe your model architecture choices for knowledge tracing. Why did you select this particular architecture (e.g., LSTM, Transformer, or hybrid models)? How did architectural decisions such as hidden dimension size, number of layers, and attention mechanisms affect performance? Support your discussion with experimental evidence.

3. (10%) Conduct an error analysis of your knowledge tracing model's predictions. Identify patterns in misclassified student responses and discuss potential reasons for these errors. Are there specific concepts, question types, or student behavior patterns where your model performs poorly? What insights can be gained from these errors?



Grading policy

- Kaggle (70%)
 - 30% based on the public leaderboard score and 70% based on the private leaderboard score
 - Leaderboard score consists of basic score and ranking score
 - Basic score :
 - Over strong baseline : 55
 - Over simple bassline : 40
 - Under simple baseline : 25
 - Ranking score:
 - $15 - (15/N) * (\text{ranking} - 1)$, N=numbers of people in the interval
- Report (30%)
 - 10% for each quesiton



E3 Submission

Submit your source code and report to E3 before 6/6 (Fri.) 23:59.

No late submission !

Follow the submission format or there will be a deduction of **5 points** for final project!

- Format
 - source code : final_<Group_ID>.py or other files (.py) you made
 - report : final_<Group_ID>.pdf

If you have any question about final project, please feel free to contact with TA : MinKuang Tsu
through email vctmk.cs13@nycu.edu.tw

Have Fun !



**Linear Algebra,
Calculus,
too difficult**

**Take Advanced Deep
Learning, Machine
Learning first**

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